

Vinod Tiwari

Data Scientist

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OBJECTIVE

Data Scientist with 6+ years of experience in **Machine Learning, Deep Learning, NLP, and Generative AI (LLMs, Agentic AI)**. Proficient in **Python, SQL, TensorFlow, PyTorch, AWS, and Docker**, with expertise in designing and deploying **production-level AI/ML systems**.

At **Deloitte**, led **Graph RAG LLM pipelines** and multi-agent LLM chatbots on AWS. Previously at **TCS**, delivered AI model quantization, vision-based NLP solutions, demand forecasting, and ETL pipelines.

Proven ability to optimize models, handle large-scale data, and deliver impactful AI solutions with focus on **scalability, efficiency, and innovation**.

TECHNICAL SKILLS

Programming & Scripting:	Python, SQL, C, Unix Shell Scripting
Data Science & ML:	NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, PySpark, Machine Learning, Deep Learning, NLP, Generative AI (LLMs), RAG, Agentic AI
Frameworks & MLOps:	TensorFlow, PyTorch, AWS, Docker, Kubernetes, MLflow, Git
APIs & Deployment:	REST API (FastAPI, Flask), Streamlit, Cloud Deployment, Serverless Pipelines, MCP server
Languages:	English, Hindi
Core Competencies:	Deployment & Infrastructure, Scaling & Debugging, Production-Level AI/ML Systems, Multi-Agent Orchestration, Prompt Optimization, Cross-Team Collaboration, Leadership

EXPERIENCE

Deloitte USI

JUN 2024 — Present

Data Scientist / AI Architect

- Led architecture, design, and deployment of an **Agentic AI multi-agent system** for Clinical Study Reports (CSR) for a pharmaceutical client. Utilized **LangGraph** and **MCP** to orchestrate multiple agents, optimized prompts for each agent, and managed distributed execution on AWS. Coordinated cross-functional team collaboration to ensure high performance and accuracy.
- Oversaw enterprise-grade **LLM chatbot systems**, ensuring scalability, high availability, and performance; implemented monitoring and logging to optimize runtime.
- Built a **Graph RAG LLM pipeline** with multi-agent models on AWS for automobile supply chain exploration; improved relationship mapping and retrieval efficiency.
- Developed **Talk-to-Infra agentic AI tool** to automatically resolve Mongo script errors, enhancing system reliability.
- Designed and deployed an **automated PDF data extraction pipeline** using **PyPDF**, **Plumber**, **AWS Lambda**, extracting structured data from PDFs and storing it in **RDS database**; implemented taxonomy mapping and optimized cloud resources for minimal runtime and cost.
- Designed microservices, vector embedding pipelines (**PostgreSQL VectorDB**), and automated job scheduling to streamline ingestion and retrieval workflows.
- Collaborated with product and research teams, providing AI/ML solutions aligned with business goals, including prompt engineering, agent orchestration, and pipeline optimization.

Tata Consultancy Services (TCS)

JAN 2019 — MAY 2024

AI Engineer / Data Scientist

- Model Quantization:** Optimized ML models using AMD framework, creating test suites for quantized layers; reduced inference costs while maintaining accuracy.
- Vision AI:** Delivered >95% accuracy in OCR pipelines combining Craft Detector, OpenCV preprocessing (+20% clarity), Tesseract OCR (+30% text accuracy). Fine-tuned InceptionV3, ResNet, Efficient-B0 on 15K+ fashion images achieving **98.9% classification accuracy**.
- NLP Solutions:** Designed sentiment analysis engines (VADER, TextBlob, Transformers) with improved accuracy using Word2Vec and embeddings. Led POCs with GPT-3, BERT, and LangChain-based chatbots.
- Forecasting:** Built demand forecasting models using Regression (Ridge, Lasso) and ARIMA, driving business strategy via elasticity and sales projections.
- ETL/Data Engineering:** Engineered data pipelines on Snowflake Azure, reducing processing time by **30%**. Automated daily reporting via Python, freeing analyst hours.

EDUCATION

Bachelor of Technology (B.Tech), Uttar Pradesh Technical University

2014 — 2018

Intermediate (Mathematics), A.I.T. Inter College

2010 — 2012

ACADEMIC PROJECTS & ACTIVITIES

- Presented **Dancing Robot** using C programming and Arduino microcontroller at MNNIT Allahabad Expo (2017).
- Developed **3D Printer Prototype** applying Mechatronics concepts (2017).
- Attended **IEEE Session** on Artificial Intelligence and Neural Science (2015).

AWARDS & RECOGNITION

TCS Golden Guru Gala Award , Tata Consultancy Services	2021
Recognized for excellence in educating Machine Learning and Deep Learning, with expertise in communicating algorithmic insights and Python programming using clear, visually compelling methods.	
TCS Faculty Award , Tata Consultancy Services	2021–2022
Honored twice as Subject Matter Expert (Python, Machine Learning) and featured in the TCS Yearbook (2021, 2022).	

CERTIFICATIONS & TRAINING

Machine Learning , Stanford University (Coursera, Andrew Ng)	2022
Remote Sensing and Deep Learning , Indian Space Research Organization (ISRO)	2021
Machine Learning Classification Bootcamp in Python , Udemy	2022
Python Programming (Python 3) , Udemy	2020
API Web Services , Udemy	2020